ECE 3337: Signals & Systems Analysis, WS17

1. (20 points) Draw a minimal op-amp circuit that realizes the following transfer function:

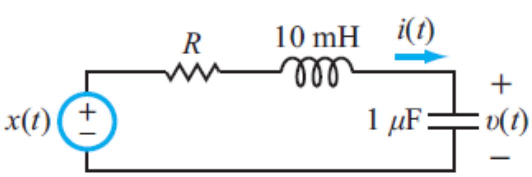
$$\mathbf{H}(\mathbf{s}) = \frac{3\mathbf{s}^2 + 4\mathbf{s} + 7}{\mathbf{s}^2 + 6\mathbf{s} + 9}$$

2. (30 points) Write down the phasor representation for each of the following signals:
   a. (5 pts) $x_1(t)$ =4sin(3t)
   b. (5 pts) $x_2(t)$ =3sin(2t-60)
   c. (5 pts) $x_3(t)$ =4cos(3t+30)
   d. (5 pts) $x_3(t)$ = sin(5t)+4cos(2t+45)
   e. (10 pts). $\frac{d(\sin(5t)+4\cos(2t+45))}{dt}$Please also express the result in the time domain.
   f. (10 pts) $\int_{0-}^{t} \sin(5\tau) + 4\cos(2\tau + 45)d\tau$. Please also ex-press the result in the time domain.

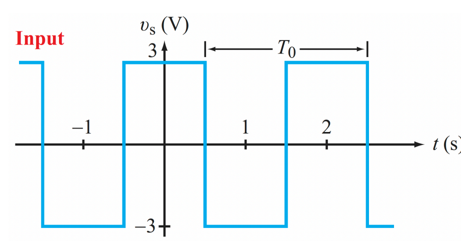
3. (20 points) Given two phasor voltages $V1 = e^{-j60}$, and $V2 = e^{j30}$, calculate the following quantities:
   a. $\mathbf{V}_1 + \mathbf{V}_2$
   b. $\mathbf{V}_1 - \mathbf{V}_2$
   c. $\mathbf{V}_1\mathbf{V}_2$
   d. $\frac{\mathbf{V}_1}{\mathbf{V}_2}$

4. (30 points) calculate the first 3 terms of the output voltage, at $(\omega_0, 3\omega_0, 5\omega_0)$ for the following RC circuit (RC = 1) when the input shown below is applied:

Input



Output



Use the Fourier series expansion in equation 5.13 of the textbook:

$$v_s(t) = \frac{12}{\pi}\left(\cos\omega_0 t - \frac{1}{3}\cos 3\omega_0 t + \frac{1}{5}\cos 5\omega_0 t - \cdots\right), \tag{5.13}$$